INFORMATICS INSTITUTE OF TECHNOLOGY

3D INTERACTIVE MEDIA DEVELOPMENT

Design and Documentation Group Report

Coursework 1 (CW 1) | Academic Year 2026/27

Module Name	3D Interactive Media Development
Module Code	5MMCS007C.1 (University of Westminster module 5MMCS007W)
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Submission Deadline	12 November 2026

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Project Title: Dehiwala-Inspired Interactive 3D Zoo Simulation
Perspective: First-Person (FPV) Camera Controller

1. Overview

Concept and Design

Research & Reference

Our project draws direct inspiration from the spatial architecture, navigational flow, and visitor layout of the National Zoological Gardens of Sri Lanka (Dehiwala Zoo). We conducted preliminary research into how physical zoo environment's structure visitor movement, balancing open-air, broad-path exploration with enclosed, controlled micro-environments (such as nocturnal houses and indoor aquariums). By studying Dehiwala Zoo's zone partitioning, we abstracted key spatial landmarks into a modular 3D virtual environment designed for intuitive real-time exploration.



Figure 1 : Dehiwala Zoo Map

Problem & Project Concept

Traditional physical zoo visits present accessibility, geographic, and financial constraints, while typical 2D static educational materials (such as photos or textbook diagrams) lack

spatial context and tactile engagement. To bridge this gap, we designed a non-game, educational 3D virtual simulation that provides an immersive, self-paced exploration experience. Users can freely navigate facility zones in first-person, inspect detailed 3D dioramas of animal enclosures, interact with informational signage, and experience localized 3D spatial audio triggers (such as species-specific calls) that mimic real-world bio-acoustic feedback.

Basic Design

The environment is constructed as an interactive 3D Diorama architectural model divided into distinct, stylized visual zones:

1. **Entrance Kiosk & Security Checkpoint:** A transitional boundary zone establishing spatial entry, complete with ticket booth assets and police/security NPC guard stations to simulate real-world arrival logic.
2. **Outdoor Terrestrial Enclosures:** Dedicated open-air zones bounded by contextual fencing, natural terrain texturing, and clear sightlines.
3. **Indoor Aquarium Structure:** An enclosed, lighting-controlled interior space utilizing localized point and spot lighting combined with atmospheric water shaders to contrast open-air daylight zones.

3DUI Theory & Concepts Used

Our system design relies on foundational 3D User Interface (3DUI) principles derived from egocentric spatial navigation and interaction theory:

1. **Egocentric First-Person Viewpoint (FPV):** Places the user's camera at eye-level height (1.7m virtual equivalent) to create natural perceptual fidelity and direct spatial presence without third-person avatar clutter.
2. **Natural Mouse-Look Spatial Orientation:** Utilizes continuous 2-axis mouse delta inputs for smooth yaw and pitch camera rotation, reducing visual disconnect and cognitive load during environmental orientation.
3. **Box Collider Boundaries for Spatial Affordance:** Employs precise BoxCollider and MeshCollider components around structural walls, perimeter fences, and terrain edges to provide physical affordance and collision feedback, preventing clip-through bugs and keeping user locomotion within valid navigational bounds.
4. **2D-in-3D UI Canvas Overlay:** Integrates a screen-space 2D overlay UI canvas for main menu controls, system pause toggles (Escape), and an interactive real-time layout map that dynamically maps spatial orientation to help users maintain spatial awareness across enclosures.

Design Diagrams

Storyboard & Flowchart

Mapped user progression from scene loading to state selection. The visual flowchart traces entry at the Main Menu Canvas -> Controls & Settings -> Spawning at the Ticket Kiosk -> Security Checkpoint Navigation -> Enclosure Pathway Traversal -> Interacting with Information Boards & Spatial Audio Triggers -> Accessing the Interactive Map / Pause State.

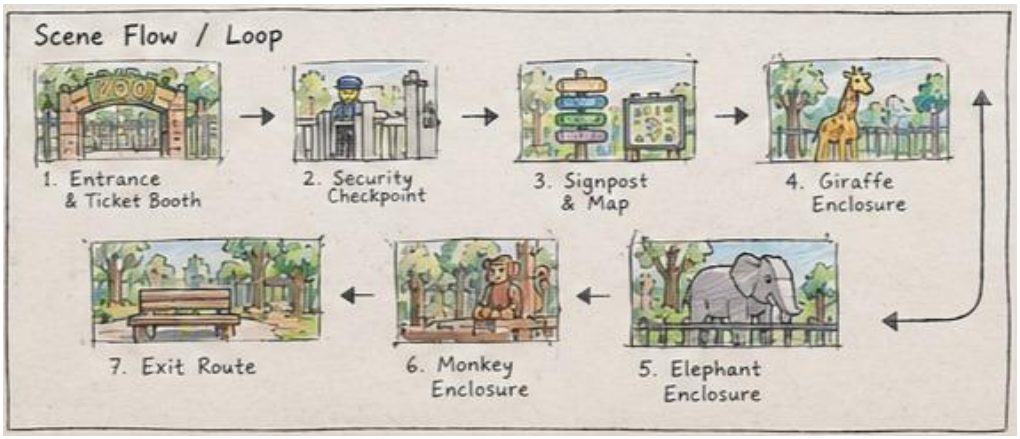


Figure 3 : StoryBoard Flow Chart

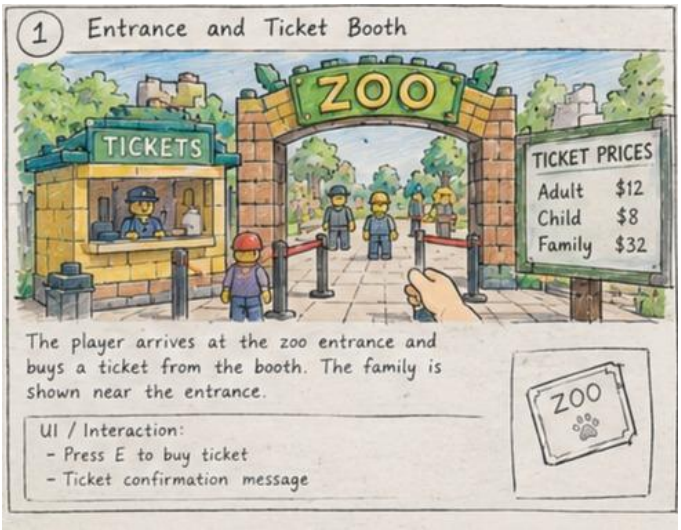


Figure 2 : Story Board (Entrance & TIcket Booth)

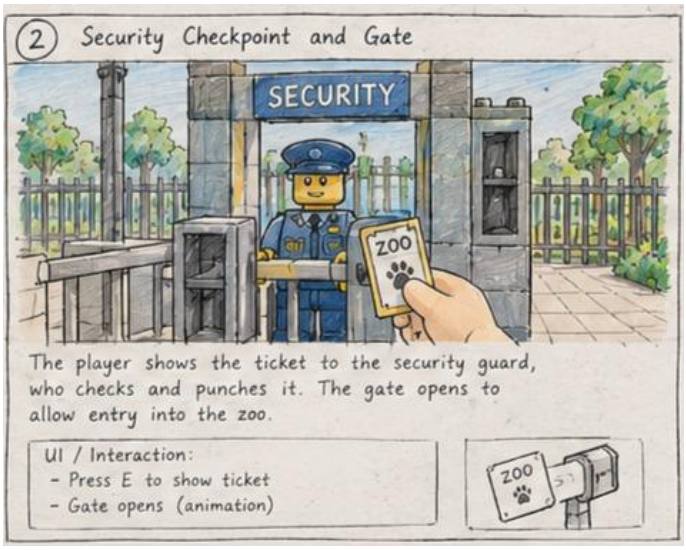


Figure 4 : Story Board (Security Checkpoint & Gate)

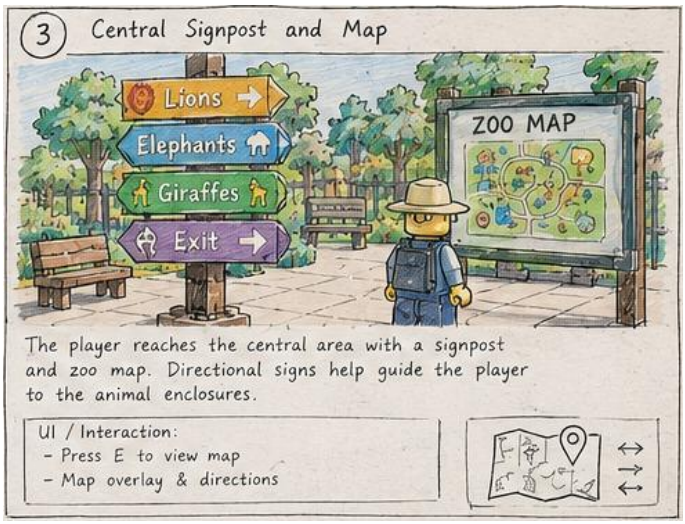


Figure 5 : Story Board (Central Signpost & Map)

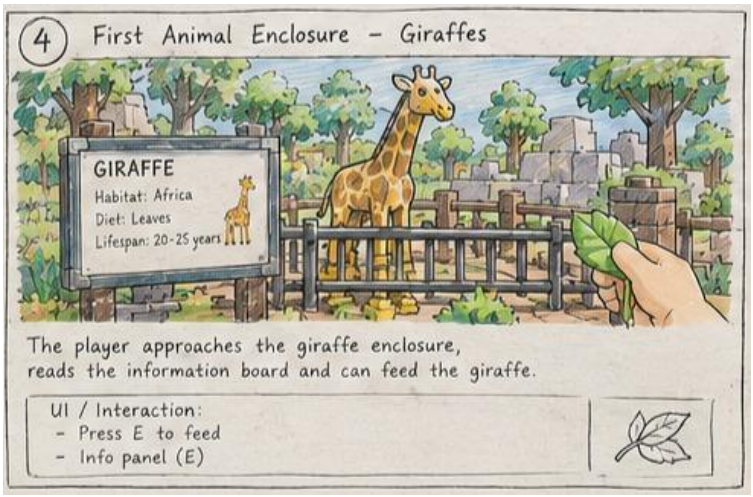


Figure 6 : Story Board (Giraffe - eg 1)

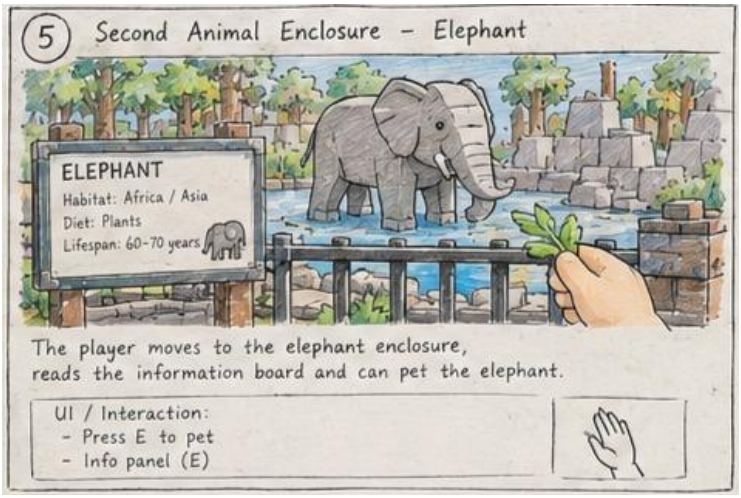


Figure 7 : Story Board (Elephant - eg 2)

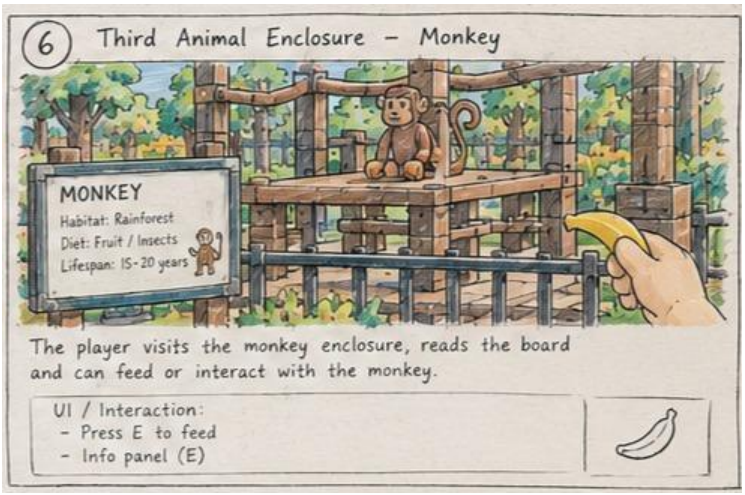


Figure 8 : Story Board (Monkey - eg 3)

Basic Use Case Diagram

Identifies primary user actors interacting with system boundary use cases: Navigate Environment (WASD), Look Around (Mouse Delta), Sprint/Jump (Shift/Space), Toggle Pause Menu (Escape), Inspect Animal Info Tag, Trigger Audio Preview, Interact with NPC (Security/Ticket), and View Real-Time Interactive Map.



Figure 9 : User Controls

Basic Domain / Class Diagram

Object-oriented architecture defining script relationships: PlayerController (manages character controller physics and movement state), CameraController (handles pitch/yaw rotation and mouse look sensitivity), UIManager (controls canvas state, pause logic, and menu sub-panels), InteractiveMapUI (syncs map markers with world coordinates), AudioManager (manages background ambience and 3D spatial audio triggers), and EnclosureInfoTrigger (detects player proximity to display species cards and play animal calls).

2. Navigation Framework

Camera & Locomotion Mechanics

Implemented a First-Person Viewpoint (FPV) controller using Unity's CharacterController component. Mouse delta inputs control camera orientation across pitch (X-axis, clamped between -80 and +80 degrees to prevent camera inversion) and yaw (Y-axis player body rotation).

Keyboard Control Scheme Rationale

1. W, A, S, D: Primary translation keys mapping directly to forward, left, backward, and right motion vectors relative to current camera heading.
2. Shift + W: Speed multiplier float applying a sprint modifier (increasing velocity from 3.5 m/s walk to 6.5 m/s run).
3. Space: Applies a vertical velocity vector based on gravity ($v_y = \sqrt{2 * g * h}$) to execute a natural jump mechanics.
4. Escape: Releases cursor lock state (CursorLockMode.None) and instantiates the in-game Pause UI canvas.

Collision Handling & Spatial Boundaries

Non-passable physical boundaries utilize non-trigger BoxCollider components attached to enclosure fencing, security gates, kiosk counters, and aquarium walls. Continuous collision detection prevents high-velocity clipping, maintaining physical affordance and spatial integrity throughout the simulation.

3. Interactivity

Current CW1 Prototype Implementation

- Functional Main Menu UI Canvas built with interactive buttons (Play, Controls, Audio/Graphics Settings, Exit).
- Esc-key focus toggle switching between locked raw input mode (CursorLockMode.Locked) and unlocked cursor menu interaction.
- Free-roam FPV navigation across all established spatial zones: Ticket Entrance, Security Checkpoint, Bear Enclosure, Duck Enclosure, Horse Enclosure, and the Aquarium interior structure.

Planned CW2 Interactivity Layer

Interactive Map Display
Real-time 2D UI map overlay accessible via shortcut key, mapping player coordinates dynamically to an overhead layout of the Dehiwala Zoo plan, displaying animal locations and facility landmarks.

Informational Boards & 3D Spatial Audio Triggers
Proximity-activated trigger volumes (OnTriggerEnter) situated in front of animal enclosures. Approaching a board instantiates a 2D HUD info card displaying species biology facts while initiating 3D spatial audio clips (AudioSource.PlayClipAtPoint) of authentic animal vocalizations.

NPC & Kiosk Service Interactions
Interactive dialogue system for ticket purchase confirmation at the entrance kiosk and security clearance interactions with police/security guard NPCs.

Animal AI & Behavior Logic
State-machine driven wildlife behaviors like roaming, idle animations, sound emissions upon approach.

4. Animation, Movement, & Narrative

Locomotion & Camera Physics
Scripted C# velocity manipulation using smooth acceleration dampening to eliminate jerky camera movement during FPV exploration.

Planned Character & Wildlife Animations
Rigged mesh keyframe and blend-tree animations for roaming animal models (bear walk cycle, duck swimming/waddling, horse idle/gallop) and looping patrol/standing idle animations for guard NPCs.

Narrative Flow & Environmental Storytelling

Structured architectural progression mirroring a real-world visit: arriving outside the facility gate, completing security and ticketing steps, and transitioning into open-ended, non-linear exploration guided by visual enclosure landmarks and the interactive map.

5. Lighting, Shaders, and Texturing

Render Pipeline Selection
Built using Unity's Universal Render Pipeline (URP) to optimize real-time shader calculations, soft shadow rendering, and performance across target hardware.

Lighting Architecture

1. **Outdoor Zones:** Single Directional Light acting as primary sunlight, paired with URP Real-Time Environment Lighting (Skybox reflection) to illuminate outdoor enclosures.
2. **Indoor Aquarium Structure:** Custom point and spot lighting fixtures with light attenuation to create a moody, deep-water aquatic ambiance.

Materials & Textures: Physically Based Rendering (PBR) materials applied across concrete pathways, wooden fencing, and glass panes, paired with a custom URP water shader featuring animated normal maps and refraction for aquarium tanks.

6. Project Management Methodology

Adopted Framework: Agile Development Methodology (Iterative Sprints).
The **Agile Development Methodology** was selected over traditional sequential models such as Waterfall due to the highly unpredictable and iterative nature of 3D virtual environment development in Unity. Building an interactive 3DUI simulation requires continuous testing of spatial mechanics, performance optimization, and asset integration, which cannot be rigidly planned upfront. Operating in 2-week iterative sprints allowed the team to construct functional increments early, such as testing first-person locomotion and camera mechanics in Sprint 1, and immediately evaluate user feel and spatial perception before committing to complex asset modeling or trigger scripting. This dynamic approach mitigated technical risks by isolating bugs early during cross-system integration and allowed flexible scope management. For instance, when collider physics caused character clipping at enclosure boundaries, the team easily re-prioritized Sprint 2 to refine `BoxCollider` boundaries and spatial affordances before integrating animal AI assets in Sprint 3. Ultimately, Agile enabled parallel workflows across C# coding, 3D asset pipelines, and UI design while ensuring continuous integration and clean version control through regular sprint reviews.

7. Commercial Analysis

Market Relevance & Future Requirements
High demand for accessible virtual educational platforms, remote museum/zoo tours, and interactive visual aids for academic institutions.

Commercial Deployment Potential
The application architecture is optimized for cross-platform export. It can be compiled as a lightweight WebGL application for embedded browser exploration on educational websites or packaged as a standalone desktop application for interactive museum kiosks.

8. AI Code Assist

- **Tool Utilized:** GitHub Copilot.

Responsible Use & Enhanced Learning
Copilot was used to accelerate repetitive C# code writing (such as boilerplate UI button event bindings, vector mathematical clamping for mouse look, and basic trigger volume syntax).

Critique, Tradeoffs & Code Review
While Copilot accelerated script generation, code generated for character movement lacked proper frame-rate independence (Time.deltaTime multipliers) and required manual refactoring. Trigger logic was manually optimized to prevent frame drops during scene state changes.

9. UI Efficacy

Conceptual Control Mapping
Mapping locomotion to WASD and spatial orientation to mouse-look leverages standard PC gaming conventions, eliminating the learning curve for users, and achieving intuitive control-to-task mapping.

Perceptual Structure & Cognitive Load
First-Person viewpoint maximizes environmental immersion by removing avatar visual clutter. The interactive overlay map provides spatial orientation on demand, preventing cognitive overload and spatial disorientation within complex enclosure paths.

Motor Coordination & Sentiment
Responsive key inputs, smooth sprint transitions (Shift), and clear multi-sensory feedback create a positive, engaging user sentiment.

10. Individual Roles (CW2 Planning)

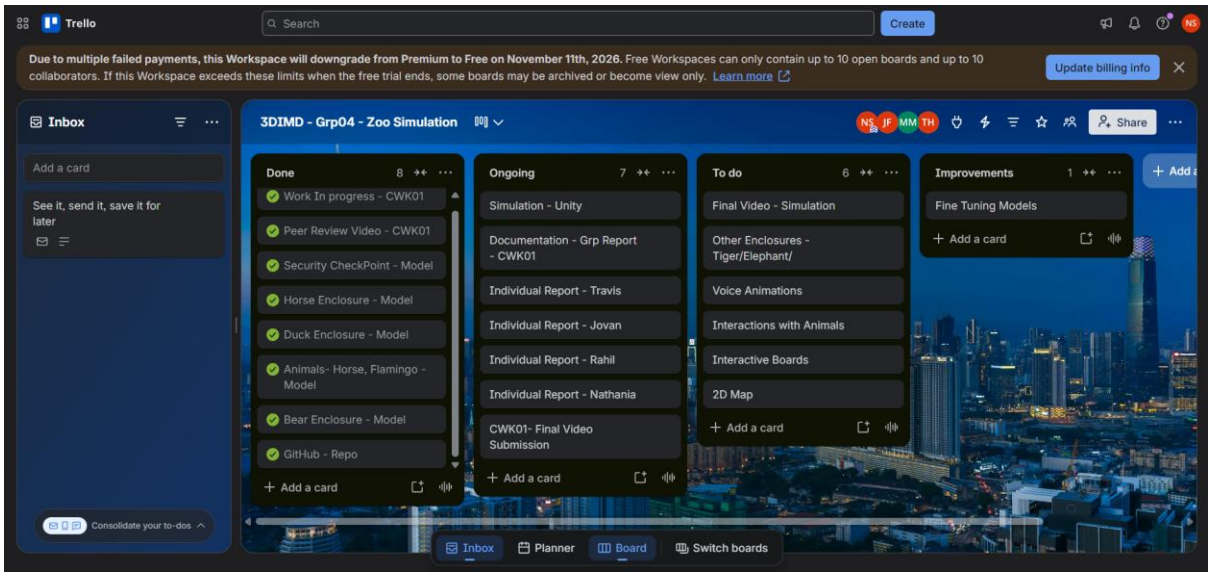


Figure 10: CW2 Planning

Team Member	Domain Area	Key Responsibilities & Deliverables
Member 1 (Nathania Sofalas)	**f. Design:** Aesthetic design and design bible **k. Coding:** Animation and trigger	Animal enclosure design & assets, animal AI animations, interaction & trigger coding, overall design bible, documentation, and GitHub repository management.
Member 2 (Travis Henry)	**j. Coding:** Movement and camera **g. Design:** Interaction design and diagrams	Security checkpoint design First-Person Controller (FPV locomotion, mouse-look camera), camera scripting, interaction diagrams, and documentation contribution.
Member 3 (Rahil Mubarak)	**b. Assets:** Asset creation **l. Animation:** Animation and Implementation	Entrance design, ticketing counter 3D asset creation, entrance animations, and ticketing C# script interactions.

Team Member	Domain Area	Key Responsibilities & Deliverables
Member 4 (Jovan Fernando)	**h. Design:** Concept and requirements **i. Coding:** Interaction	Aquarium zone concept & architectural design, aquatic asset creation, marine life swimming animations, and indoor aquarium trigger logic.